

INTRODUCTION

Background

Diffusion MRI (dMRI) provides white-matter microstructural information, while immunohistochemistry (IHC) reveals marker-specific tissue phenotypes at much finer scales. However, global or regional scalar summaries may obscure focal, boundary-localized, or region-specific patterns, motivating spatial MRI–histology association beyond direct scalar readout regression.

Challenges

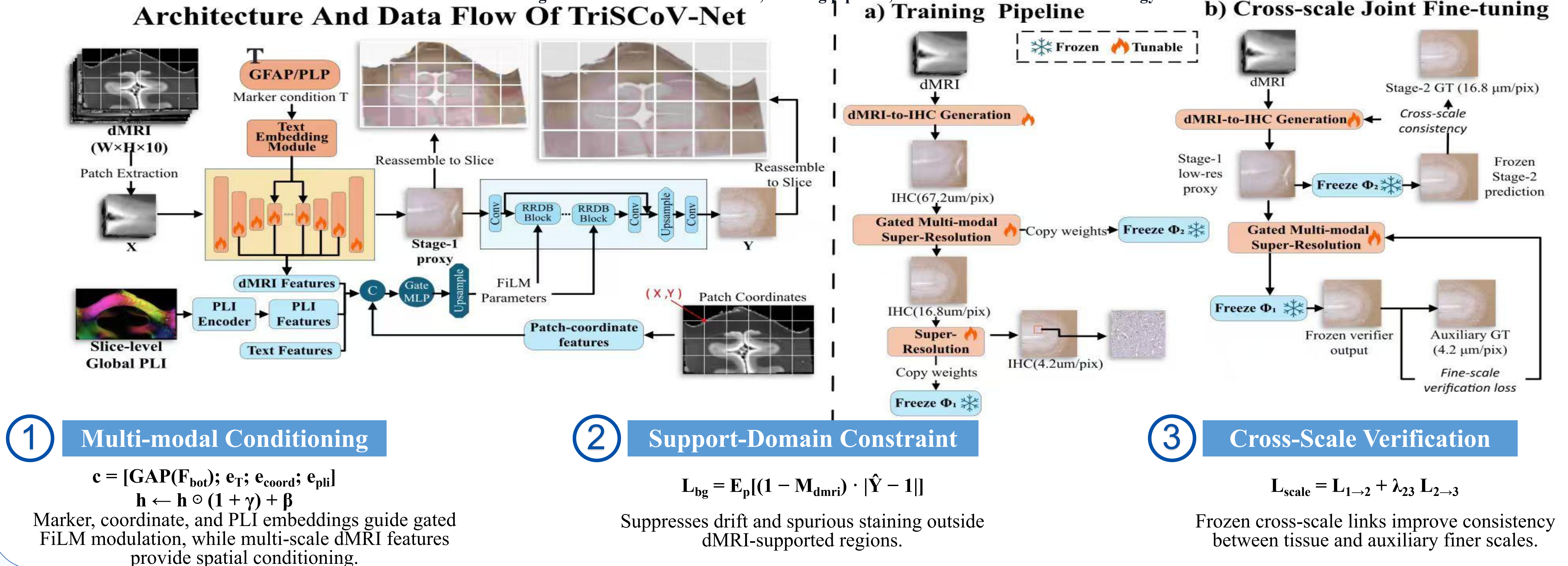
The task involves three coupled challenges: insufficient correspondence, weak-registration geometry, and cross-scale consistency. Incomplete structural and biological correspondence, deformation or partial overlap, and scale breaks may induce structural drift, spurious staining, and unstable statistics across scales.

Contributions

We propose TriSCoV-Net, a cross-scale verified framework integrating dMRI-conditioned proxy generation, PLI-assisted conditional super-resolution, support-domain constraints, and training-time cross-scale fine-tuning. The framework generates inspectable tissue-scale virtual immunomarker proxy maps that preserve regional marker distributions and support fixed-protocol quantification.

METHOD

Fig. 1. TriSCoV-Net architecture, training pipeline, and cross-scale verification strategy.



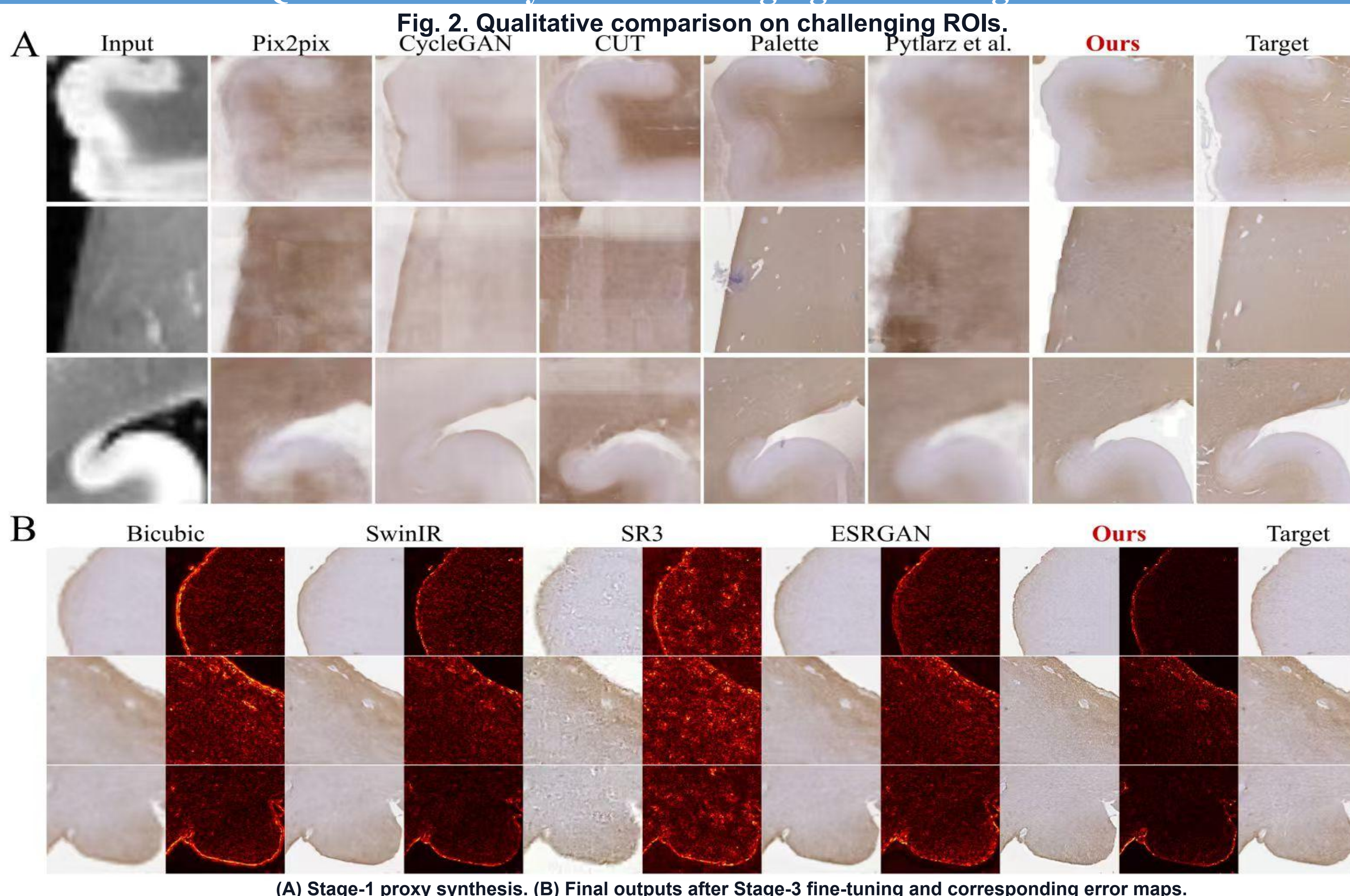
RESULTS

Image Fidelity and Tissue-Scale Quantitative Agreement

Table 1. Quantitative comparison under weak registration

Method	PSNR↑	SSIM↑	LPIPS↓	Ori.↓	Coh.↓	Area↓
Stage-1: cross-domain synthesis						
Pix2pix	15.87	0.626	0.603	41.02	0.245	14.29
CycleGAN	18.83	0.652	0.407	33.26	0.315	12.65
CUT	15.04	0.549	0.627	32.16	0.364	13.47
Palette	17.02	0.661	0.397	29.63	0.232	10.36
Pytlarz et al.	17.86	0.671	0.574	30.27	0.347	13.50
Ours	19.46	0.677	0.369	25.55	0.203	8.75
Final output after Stage-3 joint fine-tuning						
Bicubic	29.11	0.735	0.365	13.18	0.156	9.69
SwinIR	31.10	0.756	0.290	7.07	0.100	4.81
ESRGAN	30.97	0.745	0.215	7.71	0.124	6.59
SR3	26.34	0.704	0.257	18.00	0.096	6.59
Ours	31.22	0.757	0.133	4.00	0.021	4.22

Qualitative Analysis in Challenging Tissue Regions



• **Overall Performance:** Under weak registration, TriSCoV-Net yields a better balance between image consistency and fixed-protocol tissue-scale quantification. Across Stage-1 and final outputs, the method improves perceptual quality while maintaining or improving reconstruction agreement and reducing deviations from GT-based quantitative readouts.

• **Stage-1 Synthesis:** For cross-domain dMRI-to-IHC proxy generation, our method achieves the highest PSNR (19.46) and SSIM (0.677) and the lowest LPIPS (0.369) among the evaluated synthesis baselines. It also yields the lowest errors in GT-based fixed-protocol tissue-scale readouts of Orientation, Coherency, and %Area.

• **Final Output:** After Stage-3 fine-tuning, TriSCoV-Net achieves 31.22 PSNR, 0.757 SSIM, and 0.133 LPIPS, with the lowest errors in GT-based fixed-protocol tissue-scale readouts of Orientation, Coherency, and %Area.

• **Boundary Stability:** In tissue-boundary regions, representative baselines frequently exhibit background contamination, boundary spill-over, or unstable tissue contours under weak registration. TriSCoV-Net produces more conservative and spatially stable boundaries, reducing erroneous staining outside supported tissue regions.

• **Weak-Pattern Preservation:** In regions containing subtle granular expression, the final outputs after Stage-3 fine-tuning retain weak local staining patterns more clearly while maintaining a consistent background. This is particularly important because excessive smoothing can suppress focal signals that are already difficult to recover under weak alignment.

• **Low-Support Robustness:** Spurious staining is most likely to occur where dMRI support is weak or incomplete. The support-domain constraint encourages unsupported regions to revert toward background, resulting in fewer spill-over artifacts and more controllable predictions in these ambiguous areas.

• **Error Localization:** The corresponding error maps show fewer high-error hotspots and weaker boundary error bands for the proposed method. Errors are therefore more spatially concentrated, indicating reduced structural drift and improved consistency after cross-scale fine-tuning. This qualitative trend agrees with the lower LPIPS and tissue-scale quantitative errors reported in Table 1.

CONCLUSION

TriSCoV-Net introduces a dMRI-to-IHC spatial proxy-mapping framework that preserves regional marker distributions and spatial MRI–histology correspondence beyond direct scalar readout regression. By integrating dMRI-conditioned proxy generation, PLI-assisted conditional super-resolution, support-domain constraints, and training-time cross-scale fine-tuning, the framework addresses cross-domain gaps, weak alignment, and scale inconsistency. On ODBB under subject-level LOSO evaluation, TriSCoV-Net improves image fidelity and fixed-protocol tissue-scale quantitative agreement over representative baselines. The method targets quantification-oriented tissue-scale immunomarker proxies rather than non-identifiable single-cell recovery.